

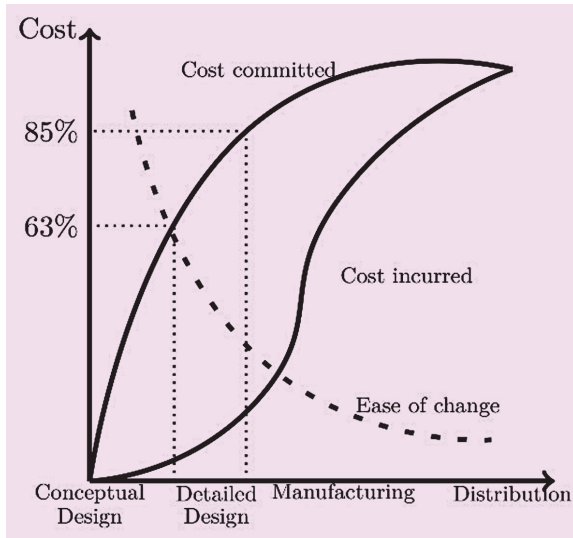


Fig. 1: Costs during different phases of product development

the whole purpose of creating variety gets defeated. Without proper training, people end up misusing the equipment. In such cases, the special feature becomes more of a hindrance than a tool.

There is a third kind of complexity, that is the operational complexity. Operational complexity arises from difficulty in operating. Certain designs require operators to perform risky or complicated procedures to make the product. We need to design our product to reduce all these three types of complexities as much as possible to make it manufacturable.

Measurement of process complexity

We need to measure things that we want to control. Without measurement, we cannot gauge our current status and determine the direction in which we are heading.

However, measuring complexity presents a challenge since it is an intangible property. We can understand when something is complex but there is no way to measure complexity directly.

To overcome this challenge, researchers have developed surrogate measures that indicate the level of complexity. There are quite a few

such indirect measures.

One such method is the complexity index (CI) calculation proposed by H.A. ElMaraghy, which is based on information theory. Instead of delving into the mathematics behind it, let's focus on how to use it in practice.

ElMaraghy's index is easy to use. It allows us to compute the CI of individual subsystems and then add them together to determine the total complexity.

We can calculate complexity as:

System Complexity: $I_s = I_1 + I_2 + \dots + I_m$, where m is the total number of subsystems

Subsystem complexity: $I_x = D_x \times H_x$

Diversity ratio: $D_x = n/N$, where n is the number of unique items in x th category and N is the number of total items in that category

Entropy: $H_x = \log_2(N + 1)$. Note that here we use log of base 2.

Let us compute the complexity of the circuit we designed in our previous article. We count the total number of each of the different types of items and their varieties.

Item	n	N	Div. Ratio	Entropy	Comp Index
ICs	3	3	1.000	2.000	2.000
Pre-assembled modules	2	2	1.000	1.585	1.585
PCBs	2	2	1.000	1.585	1.585
PCB connectors	2	2	1.000	1.585	1.585
Diodes	3	5	0.600	2.585	1.551
Resistors	4	11	0.364	3.585	1.304
Transistors	2	4	0.500	2.322	1.161
Capacitor	1	1	1.000	1.000	1.000
Buzzer	1	1	1.000	1.000	1.000
Relays	1	2	0.500	1.585	0.792
Current sensors	1	2	0.500	1.585	0.792
Total					14.355

Then we use the formulas to compute the complexity index (Table 1).

Individual CIs give us an idea of where to look for complexity reduction. Normally, there is a better scope of improvement where we have a large CI as well as a large N .

Although we have yet to design the manufacturing facilities, we will employ this process to assess the complexities of these facilities as well.

How to reduce part complexity

The best way to reduce complexity is to eliminate unnecessary parts. Eliminated parts do not need to be designed, produced, purchased, tested, stocked, assembled, and disposed of. They will never be rejected, cause production delays, become obsolete, or get lost, requiring write-off.

To determine whether a part is truly required, we should ask ourselves nine questions:

1. Does the part move relative to the others?
2. Is the movement essential for the product to function?
3. Must the part be separate for the required movement?
4. Is the part made of a different material or isolated from the rest?
5. Is this different material or

isolation essential for the product to work?

6. Must the part be separate for the required isolation or material?

7. Do we require the part to be separate for in-service adjustments or replacements?

8. Is the adjustments or replacement essential?

9. Must the part be separate for the adjustment or replacement?